Design: Advanced 3nm SoC with Low-Power Features
Technology: 3nm FinFET
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1. UNITS AND TECHNOLOGY SETTINGS

```
set_units -time ns -resistance kOhm -capacitance pF -voltage V -current mA
set_operating_conditions -max ss_0.65v_125c -min ff_0.85v_m40c
```

SOCV (Statistical OCV) Settings for 3nm

```
set_timing_derate -early 0.93 -cell_delay -net_delay
set_timing_derate -late 1.07 -cell_delay -net_delay
set_analysis_mode -analysisType onChipVariation -socv true
```

2. MASTER CLOCK DEFINITIONS (7 High-Frequency Clocks)

Clock 1: CPU Core Clock - 4.5 GHz

```
create_clock -name CPU_CLK -period 0.222 [get_ports cpu_clk_in] -waveform {0 0.111}
set_clock_transition 0.020 [get_clocks CPU_CLK]
set_clock_uncertainty -setup 0.015 [get_clocks CPU_CLK]
set_clock_uncertainty -hold 0.010 [get_clocks CPU_CLK]
set_clock_latency -source -max 0.050 [get_clocks CPU_CLK]
set_clock_latency -source -min 0.045 [get_clocks CPU_CLK]
```

Clock 2: GPU Core Clock - 3.8 GHz

```
create_clock -name GPU_CLK -period 0.263 [get_ports gpu_clk_in] -waveform {0 0.131}
set_clock_transition 0.025 [get_clocks GPU_CLK]
set_clock_uncertainty -setup 0.018 [get_clocks GPU_CLK]
set_clock_uncertainty -hold 0.012 [get_clocks GPU_CLK]
```

Clock 3: Memory Controller Clock - 4.0 GHz

```
create_clock -name MEM_CLK -period 0.250 [get_ports mem_clk_in] -waveform {0 0.125}
set_clock_transition 0.022 [get_clocks MEM_CLK]
set_clock_uncertainty -setup 0.016 [get_clocks MEM_CLK]
set_clock_uncertainty -hold 0.011 [get_clocks MEM_CLK]
```

Clock 4: AI Accelerator Clock - 3.5 GHz

```
create_clock -name AI_CLK -period 0.286 [get_ports ai_clk_in] -waveform {0 0.143}
set_clock_transition 0.028 [get_clocks AI_CLK]
set_clock_uncertainty -setup 0.020 [get_clocks AI_CLK]
```

Clock 5: NoC (Network-on-Chip) Clock - 2.5 GHz

```
create_clock -name NOC_CLK -period 0.400 [get_ports noc_clk_in] -waveform {0 0.200}
set_clock_transition 0.035 [get_clocks NOC_CLK]
set_clock_uncertainty -setup 0.025 [get_clocks NOC_CLK]
```

Clock 6: PCIe Interface Clock - 2.0 GHz

```
create_clock -name PCIE_CLK -period 0.500 [get_ports pcie_clk_in] -waveform {0 0.250}
set_clock_transition 0.040 [get_clocks PCIE_CLK]
set_clock_uncertainty -setup 0.030 [get_clocks PCIE_CLK]
```

Clock 7: Debug/Test Clock - 1.0 GHz

```
create_clock -name DBG_CLK -period 1.000 [get_ports dbg_clk_in] -waveform {0 0.500}
set_clock_transition 0.050 [get_clocks DBG_CLK]
set_clock_uncertainty -setup 0.040 [get_clocks DBG_CLK]
```

3. GENERATED CLOCKS

CPU subsystem generated clocks

```
create_generated_clock -name CPU_CLK_DIV2 -source [get_ports cpu_clk_in] \
   -divide_by 2 [get_pins cpu_pll/clk_div2]
create_generated_clock -name CPU_CLK_DIV4 -source [get_ports cpu_clk_in] \
   -divide_by 4 [get_pins cpu_pll/clk_div4]
create_generated_clock -name CPU_CLK_DIV8 -source [get_ports cpu_clk_in] \
   -divide_by 8 [get_pins cpu_pll/clk_div8]
```

GPU subsystem generated clocks

```
create_generated_clock -name GPU_CLK_DIV2 -source [get_ports gpu_clk_in] \
   -divide_by 2 [get_pins gpu_pll/clk_div2]
create_generated_clock -name GPU_SHADER_CLK -source [get_pins gpu_pll/clk_div2] \
   -multiply_by 2 [get_pins gpu_pll/shader_clk_out]
```

Memory subsystem generated clocks

```
create_generated_clock -name MEM_CLK_90 -source [get_ports mem_clk_in] \
   -edges {1 2 3} -edge_shift {0 0.0625 0.125} [get_pins mem_pll/clk_90]
create_generated_clock -name MEM_CLK_180 -source [get_ports mem_clk_in] \
   -edges {1 2 3} -edge_shift {0 0.125 0.250} [get_pins mem_pll/clk_180]
create_generated_clock -name MEM_CLK_270 -source [get_ports mem_clk_in] \
   -edges {1 2 3} -edge_shift {0 0.1875 0.375} [get_pins mem_pll/clk_270]
```

```
AI accelerator generated clocks
create_generated_clock -name AI_CORE_CLK -source [get_pins ai_pll/vco_out] \
   -multiply_by 1 -divide_by 2 [get_pins ai_pll/core_clk]
create_generated_clock -name AI_MAC_CLK -source [get_pins ai_pll/vco_out] \
   -multiply_by 2 -divide_by 3 [get_pins ai_pll/mac_clk]

Clock gating generated clocks
create_generated_clock -name CPU_GATED_CLK -source [get_pins cpu_clk_gate/CK] \
   -divide_by 1 [get_pins cpu_clk_gate/ECK]
set_clock_sense -stop_propagation [get_pins cpu_clk_gate/EN]


4. CLOCK GROUPS AND DOMAIN CROSSINGS


Define clock groups for exclusive clocks
set_clock_groups -asynchronous \
   -group {CPU_CLK CPU_CLK_DIV2 CPU_CLK_DIV4 CPU_CLK_DIV8 CPU_GATED_CLK} \
   -group {GPU_CLK GPU_CLK_DIV2 GPU_SHADER_CLK} \
   -group {MEM_CLK MEM_CLK_90 MEM_CLK_180 MEM_CLK_270} \
   -group {AI_CLK AI_CORE_CLK AI_MAC_CLK} \
   -group {NOC_CLK} \
   -group {PCIE_CLK} \
   -group {DBG_CLK}

Physical exclusive for scan clocks
set_clock_groups -physically_exclusive -group {CPU_CLK} -group {SCAN_CLK}


5. INPUT/OUTPUT CONSTRAINTS


Input delays for high-speed interfaces
set_input_delay -clock CPU_CLK -max 0.050 [get_ports cpu_data_in[*]]
set_input_delay -clock CPU_CLK -min 0.020 [get_ports cpu_data_in[*]]
set_input_delay -clock MEM_CLK -max 0.040 [get_ports ddr_dq[*]]
set_input_delay -clock MEM_CLK -min 0.015 [get_ports ddr_dq[*]]

Output delays
set_output_delay -clock CPU_CLK -max 0.045 [get_ports cpu_data_out[*]]
set_output_delay -clock CPU_CLK -min 0.018 [get_ports cpu_data_out[*]]
set_output_delay -clock PCIE_CLK -max 0.060 [get_ports pcie_tx[*]]
set_output_delay -clock PCIE_CLK -min 0.025 [get_ports pcie_tx[*]]
```

Drive strength and load

```
set_driving_cell -lib_cell BUFX16_3NM -pin Y [get_ports *_clk_in]
set_load -pin_load 0.005 [get_ports {*_out[*]}]
```

6. TIMING EXCEPTIONS

False paths

```
set_false_path -from [get_clocks DBG_CLK] -to [get_clocks {CPU_CLK GPU_CLK}]
set_false_path -from [get_ports {rst_n test_mode scan_en}]
set_false_path -through [get_pins */test_si] -through [get_pins */test_so]
```

Multicycle paths for slow logic

```
set_multicycle_path 2 -setup -from [get_pins config_reg*/Q] -to [get_pins status_reg*/D]
set_multicycle_path 1 -hold -from [get_pins config_reg*/Q] -to [get_pins status_reg*/D]
```

Max delay for critical async paths

```
set_max_delay 0.200 -from [get_pins cpu_ctrl/async_req] -to [get_pins mem_ctrl/async_ack]
```

Min delay for hold-critical paths

```
set_min_delay 0.050 -from [get_pins gpu_core/hold_critical_*]
```

7. LOW-POWER CONSTRAINTS

Power domains

```
create_power_domain PD_CPU -elements {cpu_core cpu_cache}
create_power_domain PD_GPU -elements {gpu_core gpu_mem}
create_power_domain PD_ALWAYS_ON -elements {pmu clock_gen}
```

Power states

```
create_power_state PS_ACTIVE -domain PD_CPU -voltage 0.75
create_power_state PS_SLEEP -domain PD_CPU -voltage 0.60
create_power_state PS_OFF -domain PD_CPU -voltage 0.00
```

Power switches

```
create_power_switch PS_CPU -domain PD_CPU \
   -input_supply VDD -output_supply VDD_CPU \
   -control_port {sleep_cpu} -on_state {on_state input_supply {!sleep_cpu}}
```

Level shifters

```
set_level_shifter -domain PD_CPU -applies_to outputs -location parent \
   -rule both -threshold 0.1
```

Isolation cells

```
set_isolation ISO_CPU -domain PD_CPU -isolation_power_net VDD_ALWAYS_ON \
   -isolation_ground_net VSS -clamp_value 0 -applies_to outputs
```

Retention registers

```
set_retention RET_CPU -domain PD_CPU -retention_power_net VDD_RET \
   -retention_ground_net VSS
```

Always-on buffers for critical paths

```
set_always_on_strategy AO_CRITICAL -cells {BUFAONX4_3NM INVAONX4_3NM}
apply_always_on_strategy AO_CRITICAL -to [get_nets critical_ctrl_nets*]
```

8. DRV CONSTRAINTS FOR 3NM

Maximum transition times (very tight for 3nm)

```
set_max_transition 0.030 [current_design]
set_max_transition 0.025 [get_clocks *]
set_max_transition 0.020 [get_ports *_clk_in]
```

Maximum capacitance

```
set_max_capacitance 0.040 [current_design]
set_max_capacitance 0.030 [get_nets -hierarchical *critical*]
```

Maximum fanout

```
set_max_fanout 16 [current_design]
set_max_fanout 8 [get_nets -hierarchical *high_speed*]
```

9. AREA AND POWER CONSTRAINTS

Area constraint

```
set_max_area 2500000
```

Dynamic power constraint

```
set_max_dynamic_power 15.0
```

Leakage power constraint

```
set_max_leakage_power 2.5
```

10. MMMC SETUP

Define PVT corners

```
create_corner SS_0P65V_125C -voltage 0.65 -temperature 125 -process slow
create_corner TT_0P70V_25C -voltage 0.70 -temperature 25 -process typical
create_corner FF_0P85V_M40C -voltage 0.85 -temperature -40 -process fast
```

Define modes

```
create_mode FUNC -sdc_files {func.sdc}
create_mode SCAN -sdc_files {scan.sdc}
create_mode LPWR -sdc_files {lowpower.sdc}
```

Analysis views

```
create_analysis_view av_func_ss -mode FUNC -corner SS_0P65V_125C
create_analysis_view av_func_ff -mode FUNC -corner FF_0P85V_M40C
create_analysis_view av_scan_ss -mode SCAN -corner SS_0P65V_125C
create_analysis_view av_lpwr_tt -mode LPWR -corner TT_0P70V_25C
```

Set analysis views for timing

```
set_analysis_view -setup {av_func_ss av_scan_ss av_lpwr_tt} \
            -hold {av_func_ff av_scan_ss}
```

Q)why we are taking the av_scan_ss for hold

11. SOCV ADVANCED SETTINGS

Enable SOCV with spatial correlation

```
set_socv_options -analysis_type graph_based \
            -statistical_spatial_correlation true \
            -statistical_timing true
```

SOCV derate tables for 3nm

```
read_socv_derate_table -early socv_3nm_early.table
read_socv_derate_table -late socv_3nm_late.table
```

Path-based SOCV for critical paths

```
set_timing_derate -socv_path_based -cell_delay -net_delay \
            -from [get_pins cpu_core/*] -to [get_pins cache/*]
```

12. SPECIAL 3NM CONSTRAINTS

Electromigration constraints

```
set_max_current_density 2.5 -nets {VDD* VSS*}
set_max_average_current 0.5 -pins [get_pins */VDD]
```

Via constraints for 3nm

```
set_via_enclosure_rule -min_cut 2 -cut_spacing 0.020
set_redundant_via_insertion true
```

Special routing for high-frequency nets

```
set_net_routing_rule -nets [get_nets *_CLK*] \
   -rule clock_3nm_rule -min_width 0.028 -min_spacing 0.056
```

13. USEFUL SKEW CONSTRAINTS

Enable useful skew optimization

```
set_useful_skew_options -max_allowed_skew 0.050 \
                -min_slack_improvement 0.010
```

Apply useful skew to critical paths

```
set_useful_skew -from [get_pins cpu_reg*/CK] -to [get_pins exe_reg*/D]
```

14. VERIFICATION CONSTRAINTS

Case analysis for test modes

```
set_case_analysis 0 [get_ports test_mode]
set_case_analysis 0 [get_ports scan_enable]
```

Disable timing arcs for unused modes

```
set_disable_timing [get_cells spare_cells*]
set_disable_timing -from CK -to Q [get_cells scan_flops*]
```

15. REPORT SETTINGS

Report unconstrained paths

```
set report_default_significant_digits 3
```

```
set timing_report_unconstrained_paths true
set timing_report_npaths_per_endpoint 3

 End of SDC
puts "INFO: 3nm Low-Power MMMC SDC loaded successfully"
```